Antigravity AI Internship - Remaining Development Plan

Current Status

Completed

- FastAPI Backend
- Gemini Integration
- PDF Upload Support
- DOCX Upload Support
- TXT Upload Support
- Markdown Upload Support
- Document Chunking
- Embedding Generation
- Custom Vector Store (NumPy + JSON)
- Semantic Retrieval
- RAG Pipeline
- AI Chat Interface
- Single HTML Frontend
- One-Click Startup Script
- End-to-End Working Knowledge Chatbot

This satisfies most Week 2 objectives:

- Embeddings
- Vector Search
- RAG
- Retrieval
- Response Generation
- Mini Business AI Assistant

Phase 1 — LangChain Foundations

Goal

Integrate LangChain concepts without replacing the working chatbot.

Task 1: Prompt Templates

Objective

Replace hardcoded prompts with reusable LangChain PromptTemplate objects.

Example

Current:

```
prompt = f"""
Answer using this context:

{context}

Question:
{question}
"""
```

Target:

```
PromptTemplate(
    template="""
    Use the provided context.

    Context:
    {context}

    Question:
    {question}
    """
)
```

Deliverable

- PromptTemplate integrated
- Dynamic variables supported

Task 2: Structured JSON Output

Objective

Return machine-readable responses.

Current

```
{
  "answer": "..."
}
```

Target

```
{
  "answer": "...",
  "sources": [],
  "confidence": 0.0
}
```

Deliverable

- LangChain Output Parser
- Structured JSON response

Task 3: Sequential Chain

Objective

Create multi-step processing.

Workflow

Question ↓ Retrieve Context ↓ Summarize Context ↓ Generate Final Answer

Deliverable

- First Sequential Chain implementation

Phase 2 — Conversation Memory

Goal

Convert chatbot from stateless to conversational.

Task 4: Conversation Buffer Memory

Features

- Remember previous questions
- Remember previous answers
- Support follow-up questions

Example

User: Who is Virat Kohli?

User: Which team does he play for?

Bot should understand the second question.

Deliverable

- Multi-turn conversation support

Task 5: Chat History UI

Features

- Display previous messages
- Store session history

Deliverable

- Persistent conversation window

Phase 3 — Hallucination Reduction

Goal

Improve reliability and trustworthiness.

Task 6: Source Attribution

Features

Display retrieved document chunks used for answering.

Example:

Sources:

- handbook.pdf
- policy.docx

Deliverable

- Source display below every answer

Task 7: Confidence Threshold

Features

If similarity score is low:

"I could not find relevant information in the uploaded documents."

instead of hallucinating.

Deliverable

- Refusal mechanism

Task 8: Prompt Injection Protection

Features

Ignore malicious instructions inside uploaded documents.

Example:

"Ignore previous instructions"

should not affect chatbot behavior.

Deliverable

- Basic prompt-injection guardrail

Phase 4 — Agent Features

Goal

Transform chatbot into an AI Agent.

Task 9: Calculator Tool

Features

Agent can solve calculations.

Example:

Calculate: 45 × 78

Deliverable

- Calculator Tool

Task 10: Date-Time Tool

Features

Answer:

- Today's date
- Current time

Deliverable

- Date Tool

Task 11: Agent Decision Layer

Workflow

User Query ↓ Agent ↓ Choose Tool ↓ Return Result

Examples:

Math Question → Calculator

Document Question → RAG Retrieval

Date Question → Date Tool

Deliverable

- First Working Agent

Phase 5 — LangChain RAG Migration

Goal

Learn LangChain's official RAG stack.

Do NOT remove current implementation.

Create separate experimental branch.

Task 12: LangChain Document Loaders

Implement:

- PDF Loader
- DOCX Loader

Deliverable

- LangChain ingestion pipeline

Task 13: LangChain Text Splitters

Use:

- RecursiveCharacterTextSplitter

Deliverable

- Standardized chunking

Task 14: RetrievalQA Chain

Implement:

Question ↓ Retriever ↓ LLM ↓ Answer

Deliverable

- RetrievalQA working

Task 15: LCEL Refactor

Implement:

```
chain = (
    prompt
    | model
    | parser
)
```

Deliverable

- LCEL pipeline

Phase 6 — Production Features

Task 16: Logging

Create:

logs/

- requests.log
- uploads.log
- errors.log

Deliverable

- Request logging
- Error logging

Task 17: Error Handling

Handle:

- Invalid files
- Empty uploads
- API failures
- Embedding failures

Deliverable

- Stable production behavior

Task 18: Streaming Responses

Optional Enhancement

Features

Display answer token-by-token.

Deliverable

- Typing effect

Final Internship POC

Project Title

Antigravity Knowledge Assistant Agent

Final Architecture

User ↓ Frontend ↓ FastAPI ↓ LangChain Agent ↓ ├── Memory ├── RAG Retriever ├── Calculator Tool ├── Date Tool └── Prompt Templates ↓ Gemini LLM ↓ Structured JSON Response

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Final Feature Checklist

Core AI

- [ ] Prompt Templates
- [ ] Sequential Chains
- [ ] Output Parsers
- [ ] Structured JSON
- [ ] Conversation Memory

RAG

- [x] Document Upload
- [x] Chunking
- [x] Embeddings
- [x] Retrieval
- [x] Vector Store
- [ ] Source Attribution
- [ ] Confidence Threshold

Agentic AI

- [ ] Calculator Tool
- [ ] Date Tool
- [ ] Tool Routing
- [ ] Agent Executor

Production

- [ ] Logging
- [ ] Error Handling
- [ ] Streaming Responses

Presentation Outcome

By the end of this plan, the project demonstrates:

- Prompt Engineering
- Embeddings
- Vector Search
- RAG
- LangChain
- Memory
- Agents
- Tools
- AI Safety

- FastAPI
- Gemini APIs
- Production AI Application Design

This fully covers the internship roadmap and provides a strong Semester 7 internship presentation project.